Cardiovascular System (CARD) · NPLEX-P1-CARD-02

CV-2 Histology of the Heart and Blood Vessels

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-02-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the heart wall and the fibrous skeleton

A competent response contains

· From the cavity outward: endocardium with its subendocardial layer, myocardium, epicardium.

· The endocardium is simple squamous endothelium on thin connective tissue, continuous with the endothelium of the great vessels.

· The subendocardial layer carries small vessels, nerves, and the Purkinje fibers.

· The myocardium is cardiac muscle, and the epicardium is the visceral serous pericardium: mesothelium over fat carrying the coronary vessels and nerves.

· The heart layers match the vessel layers: endocardium with intima, myocardium with media, epicardium with adventitia, and the fibrous skeleton has no vascular counterpart.

· Thickness follows chamber pressure: atria a few millimetres, right ventricle about 0.3 to 0.5 cm, left ventricle about 1.3 to 1.5 cm.

· The fibrous skeleton is dense irregular connective tissue, mainly type I collagen: four fibrous rings around the valve orifices, the right and left fibrous trigones, and the membranous interventricular septum.

· The right fibrous trigone is the thickest part and is called the central fibrous body.

· The bundle of His pierces the central fibrous body and then runs along the membranous septum, and it is the only normal electrical link between atria and ventricles.

· The skeleton anchors the valves and the myocardial sheets and completely separates atrial from ventricular muscle.

· A valve leaflet is a fold of endocardium around an essentially avascular core: a collagenous fibrosa continuous with the fibrous ring bears the load, a proteoglycan rich spongiosa cushions flexion, and an elastic layer lies on the inflow side.

· The spongiosa expands in myxomatous mitral valve disease, which is mitral valve prolapse, and vessels growing into scarred leaflets mark chronic rheumatic disease.

Distinctions that matter

· The endocardium is the layer; the subendocardium is only its deepest part, and that is where the Purkinje fibers and the ischemic vulnerability sit.

· The fibrous rings are mechanical anchors and electrical insulators at the same time, and students usually remember only the first job.

Common misconceptions

· Describing valve leaflets as vascular tissue. The normal leaflet core is essentially avascular, and vessels within it are a sign of old rheumatic scarring.

· Giving the right and left ventricle the same wall thickness. The left is about three to four times thicker because it generates systemic pressure.

Where it is taught: Section C01, paragraphs on endocardium and valves, myocardium, epicardium, and the fibrous cardiac skeleton of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C01-BD2

Brain dump 2 · integration and application: Wall stress, hypertrophy, and two failures of the skeleton

A competent response contains

· By the law of Laplace, wall stress is proportional to pressure times radius divided by twice the wall thickness.

· Case 1: (120 x 2.5) divided by (2 x 1.0) is 150 units, the baseline load on each myocyte.

· Case 2: (200 x 2.5) divided by (2 x 1.0) is 250 units, so raising pressure alone raises the load on every myocyte sharply.

· Thickening the wall to about 1.7 cm returns stress to about 147 units, which is the purpose of hypertrophy.

· Case 3: (120 x 3.5) divided by (2 x 0.8) is about 263 units, so dilation without matching thickening raises the energy cost of every beat even at a normal pressure.

· Pressure overload from hypertension or aortic stenosis adds sarcomeres in parallel, giving concentric hypertrophy with a thick wall and an unchanged or smaller cavity.

· Volume overload from regurgitation adds sarcomeres in series, giving eccentric hypertrophy with a larger cavity.

· Hypertrophied myocytes have large, dark, rectangular boxcar nuclei.

· The cost of thickening is a stiffer wall that outgrows its capillary supply, which predisposes to subendocardial ischemia.

· A strand of myocardium crossing a fibrous ring is an accessory pathway, the bundle of Kent, which bypasses the AV node and its delay.

· The three tracing findings that follow are a short PR interval, a delta wave, and reentrant tachycardia, the Wolff-Parkinson-White pattern.

· The His bundle lies beside the aortic ring and the membranous septum, so an aortic ring abscess in infective endocarditis, ring calcification, or surgery in that area can cause new AV block.

Distinctions that matter

· Concentric and eccentric hypertrophy differ in where the sarcomeres are added, which is why one thickens the wall and the other enlarges the cavity.

· Radius and thickness pull in opposite directions in the same relationship, so a dilated thin ventricle can carry a higher myocyte load than a hypertensive thick one.

Common misconceptions

· Treating hypertrophy as damage in itself. It is a response that returns wall stress toward normal, and its costs are stiffness and an outgrown capillary supply.

· Assuming an accessory pathway is an extra piece of conduction tissue. It is ordinary myocardium left crossing the ring, which is why it has no nodal delay.

Where it is taught: Section C01, myocardium paragraph, the worked example on wall stress, and the final paragraph of the section of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C01-RT01

Retrieval exercise: Rebuild why the subendocardium is injured first

A competent response contains

· Step 1: the coronary arteries run in epicardial fat and penetrate inward, so the inner wall is supplied last.

· Step 2: systole raises intramural pressure most near the cavity.

· Step 3: the inner wall vessels are compressed, so flow to the inner wall is mainly diastolic.

· Step 4: wall stress and oxygen demand are highest in the inner layers.

· Step 5: hypotension, tachycardia, or hypertrophy drops supply below demand.

· Step 6: necrosis appears first in the inner third of the wall.

· Tachycardia matters because it shortens diastole, which is when the inner wall is perfused.

· Wall stress is highest at the inner layers because stress rises with radius and the inner fibres sit at the largest effective radius for a given thickness of overlying muscle.

· The result is a subendocardial infarct, which can occur without complete occlusion of an artery because the inner third is already at the edge of its supply.

· The subendocardium is the watershed of the ventricular wall.

Distinctions that matter

· Epicardial coronary disease is about the vessel; subendocardial vulnerability is about the layer, and the two can produce ischemia independently.

· A subendocardial infarct reflects supply falling below demand across a whole layer, while a transmural infarct follows occlusion of a single artery.

Common misconceptions

· Expecting the layer nearest the blood in the cavity to be the best supplied. Cavity blood does not perfuse the wall; the coronary arteries do, from outside inward.

· Leaving systolic compression out of the explanation. Without it there is no reason the inner wall depends on diastole.

Where it is taught: Section C01, flow diagram on why the subendocardium is injured first of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C01-CA01

Clinical application: Chest tightness on exertion in a man with a thick left ventricle

A competent response contains

· Recognize: concentric hypertrophy from pressure overload, here aortic stenosis with a slow low volume carotid upstroke and a systolic murmur, with sarcomeres added in parallel.

· Explain: at a radius of 2.4 cm and 210 mm Hg, a 1.0 cm wall gives (210 x 2.4) divided by 2, which is 252 units, and a 1.8 cm wall gives (210 x 2.4) divided by 3.6, which is about 140 units, so thickening brought the load on each myocyte back toward normal.

· Differentiate: a plaque limits flow in a named epicardial artery, while this heart has smooth arteries and an oxygen demand that the supply cannot meet because the wall is thick, stiff, and has outgrown its capillary bed; the angiogram showing no narrowing is what separates them.

· Investigate: the subendocardium is at greatest risk, and on a stained section look for enlarged myocytes with large dark rectangular boxcar nuclei, the ratio of capillaries to enlarged myocytes, and interstitial fibrosis or myocyte loss concentrated in the inner third.

· Decide: necrosis would appear first in the inner third of the left ventricular wall, giving a subendocardial infarct.

· Justify: the coronary arteries run in epicardial fat and penetrate inward, systole compresses the inner vessels so inner flow is mainly diastolic, and wall stress and oxygen demand are highest in the inner layers, so the inner third is where supply fails first even with patent arteries.

Distinctions that matter

· Ischemia from demand outstripping supply across a layer is not the same as ischemia from a narrowed artery, even though the symptom is the same.

· Hypertrophy lowers wall stress while raising total oxygen demand, so the adaptation and the vulnerability come from the same change.

Common misconceptions

· Assuming exertional chest tightness always means an obstructive plaque. A thick ventricle with smooth arteries can be ischemic on demand alone.

· Expecting the outer wall to be injured first because it is furthest from the cavity. Supply arrives from the outside, so the inner third is the watershed.

Where it is taught: Section C01, myocardium paragraph, the worked example on wall stress, and the subendocardial flow diagram of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw a cardiac myocyte and its intercalated disc

A competent response contains

· A ventricular myocyte is a short branched cylinder, about 10 to 20 micrometres wide and 80 to 100 micrometres long, with one or two central nuclei.

· The clear perinuclear zone holds the Golgi apparatus, glycogen, and lipofuscin, a wear and tear pigment that accumulates with age.

· Mitochondria fill about a third or more of the cell volume, with close to one capillary per myocyte.

· Nearly all ATP comes from oxidative phosphorylation, mainly of fatty acids.

· Contraction stops within about a minute of complete ischemia, and irreversible injury begins after roughly 20 to 40 minutes.

· Myocytes join end to end at intercalated discs, dark stepped lines across the fibre; the transverse portion sits at the level of a Z line and holds the mechanical junctions, and the lateral portion holds most of the gap junctions.

· Fascia adherens: cadherins linked to the actin of the last sarcomere; it transmits contractile pull to the next cell and replaces the Z line at the cell end.

· Desmosome: desmosomal cadherins linked to desmin through plakoglobin, plakophilin, and desmoplakin; it keeps cells from pulling apart under repeated stress.

· Gap junction: connexons of connexin 43 in ventricular muscle; it passes ions between cells and makes the myocardium a functional syncytium.

· The T tubule pairs with one terminal cistern of sarcoplasmic reticulum, which is a diad, and it sits at the Z line.

· Cardiac T tubules are wide, and atrial myocytes have few of them.

Distinctions that matter

· Two junctions of the disc couple cells mechanically and one couples them electrically, and they sit in different portions of the disc.

· A diad has one cistern at the Z line, while a skeletal triad has two cisternae at the A and I band junction.

Common misconceptions

· Writing that the intercalated disc is a single junction. It is three junction types with separate proteins and separate jobs.

· Placing the nuclei at the periphery of the cell. Cardiac nuclei are central and few; peripheral and numerous nuclei belong to skeletal fibres.

Where it is taught: Section C02, paragraphs on the myocyte and intercalated discs, and the table of disc junctions of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C02-BD2

Brain dump 2 · integration and application: When the junctions fail

A competent response contains

· Heart 1: the cells stay attached because adhesion depends on the desmosomes and fascia adherens, but excitation spreads slowly and irregularly because current passes between myocytes mainly through gap junctions, which is a substrate for reentrant arrhythmia.

· Heart 2: the cells pull apart under stretch because the desmosomes hold them together; the lost myocytes are replaced by fibrofatty tissue, mostly in the right ventricle, and the result is ventricular arrhythmia and sudden death in young people.

· Heart 2 is arrhythmogenic cardiomyopathy, caused by mutations in desmosomal proteins such as plakophilin 2 or desmoplakin.

· The recessive form is Naxos disease, from a plakoglobin defect, which adds woolly hair and thickened palm and sole skin because the same desmosomal protein holds the epidermis together.

· Heart 3: adhesion is normal, but gap junctions shifted to the lateral borders slow conduction and change its direction, since conduction is normally faster along the fibre axis than across it; this occurs at the border of an old infarct and in a failing ventricle and creates a substrate for reentry.

· Adult cardiomyocytes are essentially postmitotic and have no satellite cell reserve.

· The heart therefore adapts to overload by hypertrophy and heals an infarct with collagen scar rather than new muscle.

Distinctions that matter

· Mechanical failure and electrical failure follow different junctions: desmosomes and fascia adherens for adhesion, gap junctions for current.

· Losing gap junctions and relocating them produce the same conduction problem by different routes, which is why both are arrhythmogenic.

Common misconceptions

· Expecting cells to separate when connexin 43 falls. Gap junctions pass current and contribute little to holding cells together.

· Explaining the hair and skin findings in Naxos disease as unrelated. The same desmosomal protein is used in the epidermis.

Where it is taught: Section C02, paragraph on gap junctions and desmosomal disease, the calcium handling paragraph, and the stop and predict box of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C02-CP01

Comparison table: Cardiac muscle against skeletal muscle

A competent response contains

· Cell shape and nuclei: cardiac cells are short and branched with one or two central nuclei; skeletal fibres are long and unbranched with many peripheral nuclei.

· Coupling: cardiac cells join at intercalated discs and behave as a functional syncytium; skeletal fibres have no junctions between them and each has its own motor ending.

· T tubules: cardiac tubules are wide, sit at the Z line, and pair with one cistern as a diad; skeletal tubules are narrow, sit at the A and I band junction, and pair with two cisternae as a triad.

· Trigger for sarcoplasmic reticulum release: in cardiac muscle calcium entering through L-type channels opens ryanodine receptor 2, which is calcium induced calcium release; in skeletal muscle the voltage sensor is mechanically linked to ryanodine receptor 1.

· Extracellular calcium: required in cardiac muscle and not required in skeletal muscle.

· Tetanus: not possible in cardiac muscle because of the long refractory period, and possible in skeletal muscle.

· Grading of force: cardiac force is graded by how much calcium enters and how loaded the sarcoplasmic reticulum is, since more cells cannot be recruited; skeletal force is graded by recruiting more motor units and by firing rate.

· Regeneration: minimal in cardiac muscle, which heals by scar; skeletal muscle has satellite cells that allow repair.

· What the cardiac version achieves: contraction as one unit, a chamber that cannot fuse into sustained contraction and so always refills, graded force without recruitment, and a cell built to work continuously for a lifetime.

Distinctions that matter

· Both muscle types are striated with the same sarcomere, so every difference in the table is about coordination, calcium supply, or repair, not about the contractile machinery.

· The diad against the triad is a structural difference; needing extracellular calcium is the functional consequence that follows from it.

Common misconceptions

· Saying the heart recruits more fibres to contract harder. There is no recruitment in a syncytium; force is graded by calcium.

· Treating the inability to tetanize as a weakness. It is what guarantees the ventricle relaxes and refills between beats.

Where it is taught: Section C02, calcium handling paragraph and the table comparing cardiac with skeletal muscle of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C02-Q01

Board-style question: Sudden death in a young athlete with a fibrofatty right ventricle

Correct answer: C. Plakoglobin of the desmosome

Why it is right: Plakoglobin is a desmosomal protein that links desmosomal cadherins to desmin, and the recessive defect, Naxos disease, lets myocytes detach under stretch so they are replaced by fibrofatty tissue in the right ventricle, while the same protein in epidermal desmosomes accounts for the woolly hair and the thickened palm and sole skin.

A is wrong: Loss of gap junction connexons slows and disorganizes conduction but leaves adhesion intact, so myocytes would not be pulled apart and there would be no fibrofatty replacement or skin and hair findings.

B is wrong: A defective cadherin to actin link in the fascia adherens would impair transmission of contractile force between cells, and it is not the junction shared with the epidermis that explains the hair and skin.

D is wrong: A ryanodine receptor 2 defect causes catecholamine triggered arrhythmia through disordered calcium release, with a structurally normal ventricle and no fibrofatty replacement or skin findings.

A competent response contains

· Myocytes pulled apart with fibrofatty replacement, mostly of the right ventricle, with ventricular arrhythmia and sudden death in a young person, is arrhythmogenic cardiomyopathy from a desmosomal protein defect.

· Woolly hair with thickened palm and sole skin identifies the recessive plakoglobin form, Naxos disease, because plakoglobin is also used by epidermal desmosomes.

Distinctions that matter

· Desmosomal failure separates cells mechanically; gap junction failure slows conduction without separating them.

Common misconceptions

· Reading any inherited ventricular arrhythmia as an ion channel defect. Here the histology shows cells pulled apart, which points to adhesion.

Where it is taught: Section C02 of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C03-BD1

Brain dump 1 · foundational retrieval: Purkinje fibers, working myocytes, and nodal cells

A competent response contains

· Purkinje fibers form a network in the subendocardium of both ventricles and are continuous with the bundle branches.

· They are larger than working myocytes, with pale glycogen rich cytoplasm that is periodic acid Schiff positive, a few myofibrils pushed to the periphery, and one or two central nuclei often ringed by a clear halo.

· Three features make them fast: a large diameter that lowers internal resistance to longitudinal current, abundant gap junctions that lower resistance between cells, and a large fast sodium current driving depolarization.

· Their conduction velocity is about 1.5 to 4 m/s, the fastest in the heart, so the ventricles depolarize almost together and the normal QRS is narrow.

· Table rows: size and staining, largest and pale and PAS positive for Purkinje, intermediate and eosinophilic and densely striated for the working myocyte, smallest and pale in dense connective tissue for the nodal cell.

· Myofibrils: few and peripheral in Purkinje fibers, abundant in working myocytes, few in nodal cells.

· Gap junctions: abundant in Purkinje fibers and working myocytes, sparse in nodal cells.

· Conduction velocity: about 1.5 to 4 m/s in Purkinje fibers, about 0.3 to 0.5 m/s in working ventricular muscle, and about 0.05 m/s in the AV node.

· Main role: rapid distribution of the impulse, force, and pacemaking with AV delay respectively.

· Purkinje fibers pace at about 20 to 40 beats per minute, which appears as a wide complex ventricular escape rhythm.

· The cerebellar Purkinje neuron is an unrelated cell that shares the name and is a common matching trap.

Distinctions that matter

· Conduction velocity and pacing rate are separate properties: Purkinje fibers conduct fastest and pace slowest.

· Pale cytoplasm has two different causes here: glycogen in the Purkinje fiber and sparse contractile material in the nodal cell.

Common misconceptions

· Expecting the fastest conducting cell to be packed with myofibrils. Purkinje fibers trade contractile machinery for speed.

· Placing Purkinje fibers in the mid myocardium. They lie in the subendocardium, which is also why surviving ones sit at the inner edge of an infarct.

Where it is taught: Section C03, Purkinje fiber paragraphs and the comparison table of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C03-BD2

Brain dump 2 · integration and application: Read two sections and follow the consequences

A competent response contains

· The first section shows Purkinje fibers, built for rapid distribution of the impulse through the ventricles.

· Their glycogen supports anaerobic glycolysis, which is thought to help them tolerate ischemia; together with diffusion from cavity blood this lets fibers at the inner edge of an infarct survive.

· Those surviving fibers can become a source of ectopic beats.

· The second section shows atrial myocytes with granules storing atrial natriuretic peptide, and the granules are most numerous in the right atrium.

· ANP is released when increased volume or pressure stretches the atrium, which makes the heart an endocrine organ.

· ANP binds natriuretic peptide receptor A, which carries its own guanylyl cyclase domain, so the second messenger is cGMP.

· Its actions: it dilates the afferent and constricts the efferent arteriole to raise GFR, inhibits collecting duct sodium reabsorption, suppresses renin and aldosterone, and relaxes vascular smooth muscle.

· Ventricular myocytes make B type natriuretic peptide, mainly on demand in response to wall stretch rather than storing it in prominent granules.

· Plasma BNP or NT-proBNP rises in heart failure because the ventricular wall is stretched, which is why it is the one measured in blood.

Distinctions that matter

· ANP is stored in atrial granules and released by stretch; BNP is made by the ventricle largely on demand, which is why only one of them shows as prominent granules on electron microscopy.

· The same cell type underlies both sections: both Purkinje fibers and ANP storing atrial cells are modified cardiomyocytes that traded contractile machinery for another job.

Common misconceptions

· Reading glycogen rich pale cells beneath the endothelium as fat or as degenerating muscle. The PAS reaction and the peripheral myofibrils identify Purkinje fibers.

· Assuming all natriuretic peptide comes from the atrium. Ventricular BNP is the one that rises in heart failure.

Where it is taught: Section C03, Purkinje glycogen paragraph, atrial granule paragraph, and the ANP receptor paragraph of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C03-RT01

Retrieval exercise: Rebuild the atrial stretch feedback loop

A competent response contains

· Step 1: blood volume or atrial pressure rises.

· Step 2: atrial myocytes stretch and release ANP from granules.

· Step 3: ANP binds natriuretic peptide receptor A and raises cGMP.

· Step 4: GFR rises while sodium reabsorption, renin, and aldosterone fall.

· Step 5: sodium and water loss lowers volume and atrial stretch.

· The releasing cells are atrial myocytes, the hormone was stored in membrane bound dense granules near the nucleus and Golgi apparatus, and the granules are most numerous in the right atrium.

· Natriuretic peptide receptor A carries its own guanylyl cyclase domain, so the second messenger is cGMP.

· At the kidney the afferent arteriole dilates and the efferent arteriole constricts, which raises glomerular filtration.

· This is negative feedback: the last step removes the stretch that released the hormone.

· ANP also relaxes vascular smooth muscle, which lowers the load against which the heart works.

Distinctions that matter

· ANP and nitric oxide both act through cGMP, but ANP uses a receptor with its own cyclase domain while nitric oxide uses soluble guanylyl cyclase inside the target cell.

· ANP raises GFR by acting on both arterioles in opposite directions, not by raising systemic pressure.

Common misconceptions

· Calling this positive feedback because the hormone rises with the stimulus. The output of the loop cancels the stimulus, which is negative feedback.

· Expecting ANP to act through cAMP by analogy with other peptide hormones. Its receptor makes cGMP directly.

Where it is taught: Section C03, cycle diagram on atrial stretch and natriuretic peptide feedback of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C04-BD1

Brain dump 1 · foundational retrieval: Draw the wall of every vessel class

A competent response contains

· All arteries and veins have an intima of endothelium and thin subendothelial tissue, bounded in arteries by the internal elastic lamina, a media of smooth muscle and the matrix it secretes, and a mainly collagenous adventitia carrying nerves and, in large vessels, the vasa vasorum.

· Elastic artery, such as the aorta, pulmonary trunk, and major branches: thick intima and a media of about 40 to 70 fenestrated elastic lamellae; it stores energy in systole and recoils in diastole.

· Muscular artery, such as the coronary, radial, and renal: a sharp wavy internal elastic lamina and a thick smooth muscle media of up to about 40 layers; it distributes flow among organs.

· Arteriole: one or two smooth muscle layers, with an elastic lamina only in the larger ones; it is the main resistance vessel and controls capillary inflow, and the cutoff is generally under about 0.1 mm.

· Capillary: endothelium and basal lamina with pericytes only, about 5 to 10 micrometres wide; it is the exchange vessel, and its huge total cross section slows flow for exchange.

· Postcapillary venule: endothelium, pericytes, and loose junctions; it is the site of leukocyte emigration and inflammatory leak.

· Medium vein: thin media with intimal folds forming valves, for low pressure return.

· Large vein: thin media with the adventitia thickest and carrying longitudinal muscle, acting as a capacitance reservoir.

· The internal elastic lamina is most distinct in muscular arteries.

· The vasa vasorum run in the adventitia and outer media and supply the part of the wall too far from the lumen to be fed by luminal diffusion; they are more plentiful in veins.

· Valves are intimal folds in limb veins and work with the skeletal muscle pump; they are absent from the venae cavae, the portal system, and the dural sinuses.

· Veins are thin walled and compliant and hold about 60% to 70% of blood volume.

Distinctions that matter

· Arteries are built around the media and large veins around the adventitia, which is the quickest way to tell a large vein from an artery on a section.

· The elastic lamellae of an elastic artery and the smooth muscle layers of a muscular artery are both in the media, but one stores energy and the other sets calibre.

Common misconceptions

· Expecting the thickest wall layer to be the media in every vessel. In a large vein the adventitia is the thickest layer.

· Treating the vasa vasorum as a minor detail. They are why syphilitic aortitis damages the outer media and why a thick wall needs its own supply.

Where it is taught: Section C04, opening paragraph, the table of vessel classes, and the paragraphs on capillaries, venules, and veins of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C04-BD2

Brain dump 2 · integration and application: Which layer is damaged, and what the pressures do

A competent response contains

· High systolic with low diastolic pressure in an older patient: the elastin of the elastic artery media has fragmented and been replaced by collagen, so the aorta cannot store and return ejection energy, giving isolated systolic hypertension.

· Fibrillin 1 defect with a dilated root and tearing pain: Marfan syndrome weakens elastin in the media, producing cystic medial degeneration, aortic root dilation, and dissection, which splits the media.

· Tertiary syphilis with a tree bark intima: obliterative endarteritis of the vasa vasorum in the adventitia makes the outer media ischemic, so the ascending aorta becomes aneurysmal.

· Pipestem calcification with no luminal narrowing: Monckeberg medial calcific sclerosis calcifies the internal elastic lamina and the media of muscular arteries.

· Thickened renal arterioles with pink material in hypertension and diabetes: hyaline arteriolosclerosis deposits leaked plasma protein in the wall, and in severe hypertension hyperplastic arteriolosclerosis adds concentric onion skin smooth muscle; both narrow the lumen and reduce renal perfusion.

· Atherosclerosis, by contrast, affects the intima of elastic and muscular arteries.

· Resistance varies with the inverse fourth power of the radius, so a 20% narrowing raises resistance about 2.4 times, since 1 divided by 0.8 to the fourth power is about 2.44.

· Pressure falls across the arterioles from about 85 mmHg to about 30 to 35 mmHg, the largest drop anywhere in the circulation, because they combine a small radius with smooth muscle that can change it.

· In systole the aorta stretches and stores ejection energy; in diastole it recoils, maintaining diastolic pressure and nearly continuous distal flow, the windkessel effect.

· Loss of elastin raises systolic pressure, lowers diastolic pressure, and widens pulse pressure.

Distinctions that matter

· Intimal disease narrows the lumen, medial disease weakens or stiffens the wall, and adventitial disease starves the outer wall, so naming the layer predicts the clinical picture.

· Monckeberg calcification and atherosclerosis both calcify, but one sits in the media without narrowing the lumen and the other sits in the intima and does narrow it.

Common misconceptions

· Reading a wide pulse pressure in an older patient as a strong heart. It reflects a stiff aorta that no longer absorbs the stroke volume.

· Assuming a small narrowing has a small effect on resistance. The fourth power relationship means a 20% narrowing more than doubles it.

Where it is taught: Section C04, paragraphs on arteries and arterioles, the flow diagram on elastic recoil, and the arteriolar paragraph of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C04-Q02

Board-style question: Capillary type and what can leave the vessel

Correct answer: A. Whole red and white blood cells

Why it is right: Large intercellular gaps together with a discontinuous basal lamina leave no barrier to cells, which is why sinusoidal capillaries in the liver, spleen, and bone marrow allow whole blood cells and large proteins to cross the wall.

B is wrong: Water and dissolved gases alone describe a continuous capillary with unbroken endothelium and an intact basal lamina, as in muscle, lung, and the nervous system.

C is wrong: Passing small hormones while holding back albumin describes a fenestrated capillary, whose pores of about 60 to 80 nm sit in an otherwise continuous wall over a continuous basal lamina.

D is wrong: A wall sealed by tight junctions describes the continuous capillaries of the blood brain barrier, which is the opposite of the discontinuous wall in this micrograph.

A competent response contains

· A wide lumen with large gaps between endothelial cells and a discontinuous basal lamina describes a sinusoidal capillary.

· Sinusoidal capillaries are found in liver, spleen, and bone marrow, and they let large proteins and whole cells pass.

· Permeability rises from continuous to fenestrated to sinusoidal.

Distinctions that matter

· Fenestrated capillaries have pores of about 60 to 80 nm within an otherwise continuous wall and basal lamina, so they pass fluid and hormones but not cells.

Common misconceptions

· Treating any gap in a capillary wall as a fenestration. Fenestrations are pores in the cell; sinusoidal gaps are spaces between cells with a broken basal lamina.

Where it is taught: Section C04 of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C05-BD1

Brain dump 1 · foundational retrieval: Build the endothelial product table and the nitric oxide pathway

A competent response contains

· Nitric oxide: soluble guanylyl cyclase and cGMP; relaxation, with inhibition of platelets and leukocyte adhesion.

· Prostacyclin: IP receptor, Gs, cAMP; relaxation and inhibition of platelet aggregation.

· Endothelin-1: ETA receptor, Gq, IP3 and calcium; strong sustained constriction and smooth muscle growth.

· von Willebrand factor: bridges collagen to platelet GPIb and carries factor VIII; platelet adhesion.

· P-selectin: binds sialyl Lewis X on leukocytes; leukocyte rolling.

· Thrombomodulin: binds thrombin, which then activates protein C; anticoagulant.

· Heparan sulfate: enhances antithrombin; anticoagulant.

· Tissue plasminogen activator: converts plasminogen to plasmin; fibrinolytic.

· Endothelial nitric oxide synthase makes nitric oxide and L-citrulline from L-arginine and oxygen, using NADPH and tetrahydrobiopterin.

· Besides relaxing smooth muscle, nitric oxide inhibits platelet adhesion, platelet aggregation, leukocyte adhesion, and smooth muscle proliferation.

· Superoxide and hemoglobin destroy nitric oxide within seconds, so its action is local.

· Weibel-Palade bodies are rod shaped endothelial granules holding von Willebrand factor with P-selectin in their membrane, released within minutes by thrombin, histamine, epinephrine, or vasopressin acting on V2 receptors.

· von Willebrand factor bridges exposed collagen to platelet glycoprotein Ib, which matters most at high shear, and carries factor VIII, so severe deficiency lowers factor VIII.

· ADAMTS13 trims ultralarge von Willebrand factor multimers, and its deficiency lets them aggregate platelets in small vessels, causing thrombotic thrombocytopenic purpura.

· Preformed P-selectin starts leukocyte rolling within minutes, whereas E-selectin is synthesized over hours after interleukin 1 or tumor necrosis factor.

Distinctions that matter

· Nitric oxide and prostacyclin both relax and both inhibit platelets, but one works through cGMP and the other through cAMP.

· P-selectin is preformed and released in minutes, while E-selectin has to be made over hours, which is why the two act at different points of an inflammatory response.

Common misconceptions

· Treating nitric oxide as a circulating signal. It survives seconds and acts on the smooth muscle immediately beneath the endothelium.

· Thinking von Willebrand factor only sticks platelets down. It also carries factor VIII, which is why a severe deficiency affects the coagulation cascade too.

Where it is taught: Section C05, nitric oxide paragraph, the paragraph on Weibel-Palade bodies, and the table of endothelial products of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C05-BD2

Brain dump 2 · integration and application: Two vessels, one drug, and the same receptor

A competent response contains

· In the intact ring, acetylcholine binds endothelial M3 receptors, raises endothelial calcium, activates eNOS, and the nitric oxide produced relaxes the underlying smooth muscle, outweighing the direct effect on muscle.

· In the denuded ring, acetylcholine reaches only smooth muscle M3 receptors, which act through Gq, IP3, and calcium to cause contraction.

· The same receptor on two different cells therefore moves the vessel in opposite directions, which is why the response is a test of the endothelium rather than of the muscle.

· Infused into a human coronary artery, acetylcholine dilates a healthy vessel but constricts one with atherosclerosis or risk factor driven dysfunction, revealing lost bioavailable nitric oxide.

· This matters because light microscopy shows structure, not function, so a normal looking endothelium can be profoundly dysfunctional and is detected by its response to flow or acetylcholine.

· Rising flow raises shear stress and releases nitric oxide, so the upstream artery widens, which is flow mediated dilation.

· Aspirin irreversibly acetylates cyclooxygenase.

· Anucleate platelets cannot replace the enzyme, while endothelial cells can resynthesize it, so a low dose shifts the balance away from platelet thromboxane A2 and toward endothelial prostacyclin.

Distinctions that matter

· The acetylcholine response reports endothelial function; the drug itself is not vasodilator or vasoconstrictor in isolation.

· Thromboxane A2 and prostacyclin come from the same enzyme in two different cells, which is what makes a dose difference possible.

Common misconceptions

· Describing acetylcholine as a vasodilator without qualification. It dilates only when a functioning endothelium is there to make nitric oxide.

· Expecting a normal looking endothelium on a section to be working normally. Function has to be tested with flow or acetylcholine.

Where it is taught: Section C05, nitric oxide paragraph, the worked example on the acetylcholine paradox, the prostacyclin paragraph, and the integration note on structure against function of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C05-RT01

Retrieval exercise: Rebuild the nitric oxide relaxation pathway

A competent response contains

· Step 1: shear stress, acetylcholine, or bradykinin act on the endothelium.

· Step 2: calcium with calmodulin, together with phosphorylation, activate eNOS.

· Step 3: nitric oxide diffuses into the underlying smooth muscle.

· Step 4: soluble guanylyl cyclase raises cGMP and protein kinase G is activated.

· Step 5: calcium falls and myosin light chain phosphatase is activated.

· Step 6: the smooth muscle relaxes and the vessel dilates.

· The same nitric oxide and cGMP signal inhibits platelets, which is why the healthy endothelium keeps blood fluid and vessels relaxed at the same time.

· Atrial natriuretic peptide raises the same second messenger, cGMP, through natriuretic peptide receptor A, which carries its own guanylyl cyclase domain.

Distinctions that matter

· Nitric oxide is made in the endothelium and acts in the smooth muscle, so the signalling steps are split between two cells.

· Relaxation here comes from both less calcium and more phosphatase activity on myosin, not from calcium alone.

Common misconceptions

· Putting the cyclase in the endothelial cell. Soluble guanylyl cyclase is in the smooth muscle cell that nitric oxide diffuses into.

· Expecting nitric oxide to use a surface receptor. It is a small gas that crosses the membrane and binds its target enzyme directly.

Where it is taught: Section C05, flow diagram on nitric oxide and vascular relaxation, with the ANP receptor in Section C03 of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C05-PR01

Prediction exercise: Predict what happens to an endothelium under oxidative stress

A competent response contains

· Superoxide rises, because smoking, diabetes, hypertension, and oxidized LDL all raise endothelial oxidative stress.

· Available nitric oxide falls, because superoxide destroys it within seconds, and reduced bioavailable nitric oxide is the hallmark of endothelial dysfunction.

· Once tetrahydrobiopterin runs low, eNOS uncouples and produces superoxide instead of nitric oxide, so the enzyme adds to the problem it normally opposes.

· Endothelin-1 rises and the vessel constricts, since nitric oxide, prostacyclin, and laminar shear normally suppress it.

· VCAM-1, ICAM-1, and the selectins rise, so monocytes adhere and enter the intima along with LDL.

· Tissue factor and plasminogen activator inhibitor 1 rise while thrombomodulin falls, so the surface becomes prothrombotic, which is the endothelial injury of Virchow's triad.

· Infused acetylcholine constricts this vessel instead of dilating it, because endothelial nitric oxide no longer outweighs the direct effect on smooth muscle.

· Lesions appear first at branch points and curves, where shear is low and disturbed rather than laminar.

· The cycle is self reinforcing: the intimal lesion generates more reactive oxygen species, which drives progression toward atherosclerosis.

Distinctions that matter

· Endothelial dysfunction is a shift in the balance of products, not a loss of the endothelium, which is why the vessel can look normal on a section.

· Low laminar shear promotes dysfunction while high laminar shear protects, so the location of plaques follows flow rather than the burden of risk factors.

Common misconceptions

· Expecting an enzyme that makes nitric oxide to simply make less of it. Uncoupled eNOS makes superoxide, so it switches sides.

· Assuming plaques form where flow is fastest. They form where shear is low and disturbed, at branch points and curves.

Where it is taught: Section C05, endothelial dysfunction paragraph and the cycle diagram, with the acetylcholine example earlier in the section of the CV-2 Histology of the Heart and Blood Vessels notes.